2-6 BRAKE CLEARANCE MEASUREMENT/ADJUSTMENT

id051700664800

Preparation Before Servicing

1. Print out the measurement/adjustment value input sheet. (See MEASUREMENT/ADJUSTMENT VALUE INPUT SHEET.)

Note

- When performing the measurement/adjustment, input the measured and calculated values into the measurement/adjustment value input sheet.
- If the measurement/adjustment value input sheet has already been printed out for the other measurements/adjustments, use the sheet.

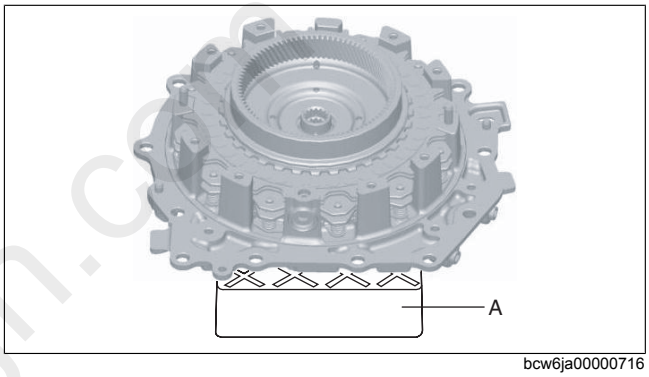
2-6 Brake Clearance Measurement

1. Set the end cover with the assembled part on the workbench as shown in the figure.

Caution

- **To reduce measurement differences during the 2-6 brake clearance measurement, use the rubber plates to adjust the alignment surface of the end cover with the transaxle case so that it is level.**

A : Rubber plate

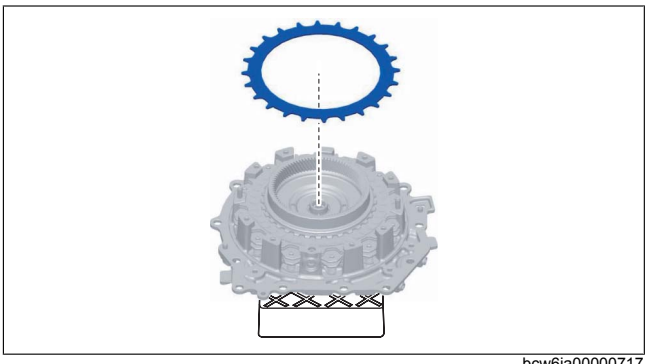


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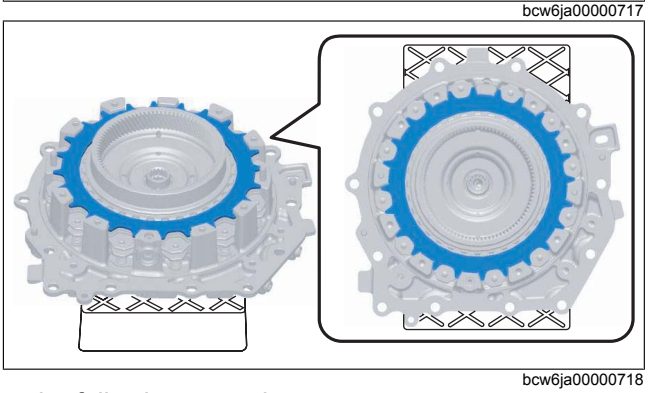
2. Assemble the retaining plate.

Note

- Retaining plate size: Inner diameter approx. 137.2 mm {5.402 in}



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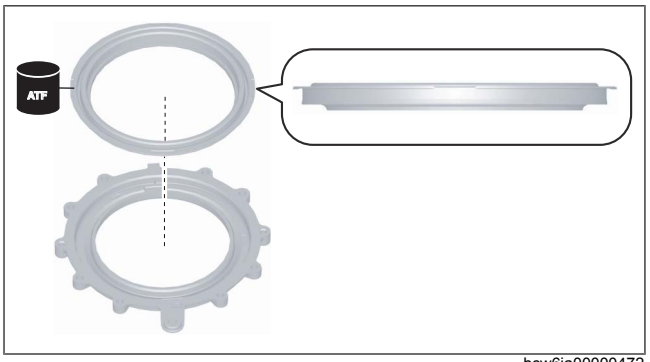


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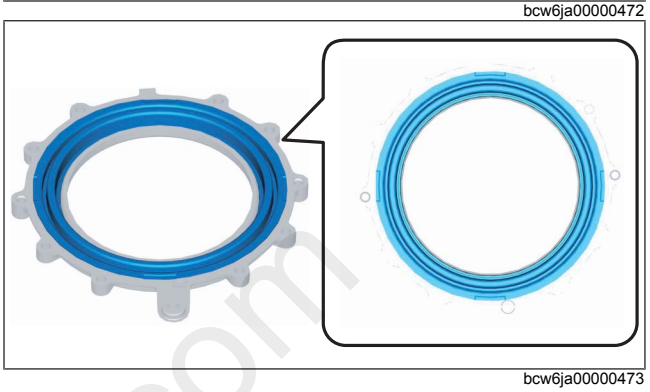
3. Assemble the 2-6 brake piston to the brake housing using the following procedure:
   (1) Apply ATF (ATF FZ) to the 2-6 brake piston lip.

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(2) Assemble the 2-6 brake piston.



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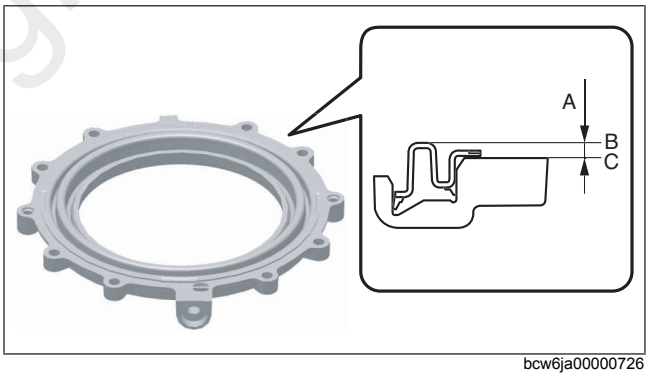
4. Measure distance A shown in the figure of the assembled parts from Step 3 and calculate the average value of distance A.

Note
- Recommended measuring instrument: Depth micrometer, straight edge ruler

A : Distance A
B : 2-6 brake piston end
C : Brake housing end (alignment surface with end cover)

(1) Measure the thickness of the straight edge ruler.

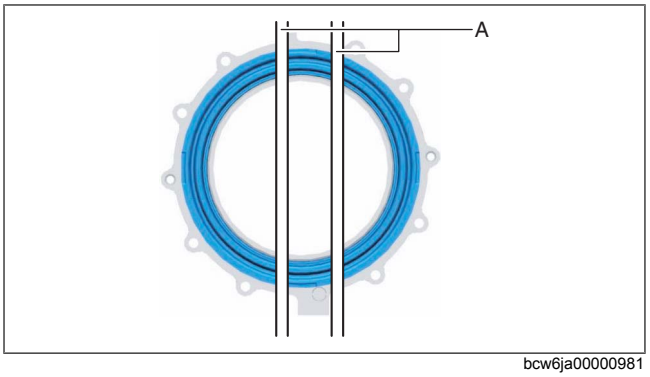
Note
- Recommended measuring instrument: Micrometer



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(2) Input the measured straight edge ruler thickness into the measurement/adjustment value input sheet.
(3) Set two straight edge rulers on the 2-6 brake piston as shown in the figure.

A : Straight edge ruler



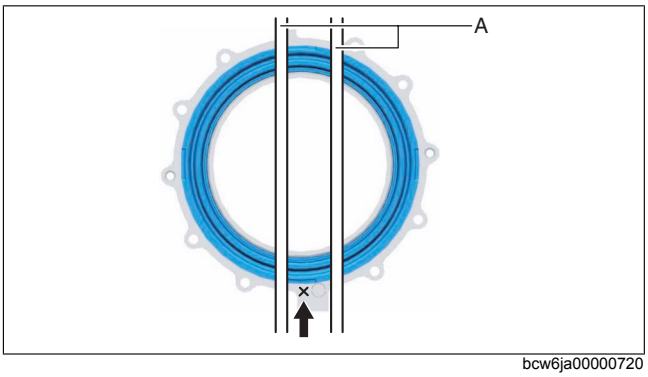
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(4) Measure the positions in 3 locations (each separated by 120°) shown in the figure and subtract the thickness of the straight edge ruler from distance A.

A : Straight edge ruler

(5) Input distance A and the average value of distance A into the measurement/adjustment value input sheet.

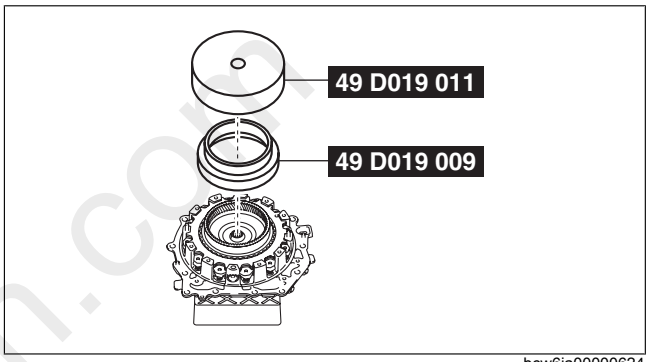


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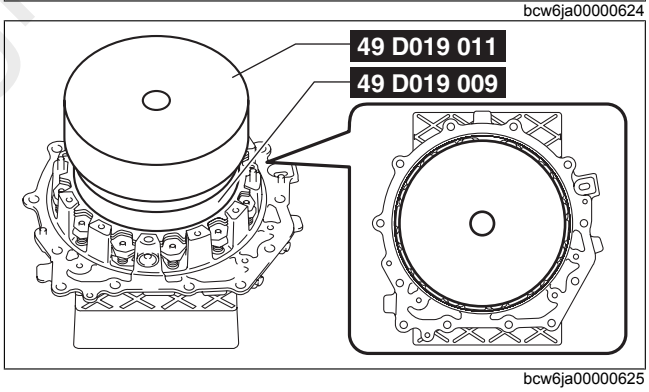
5. Install the SSTs.

Caution

- **Install the SST (49 D019 009) so that the end with the large diameter is pointing towards the retaining plate.**



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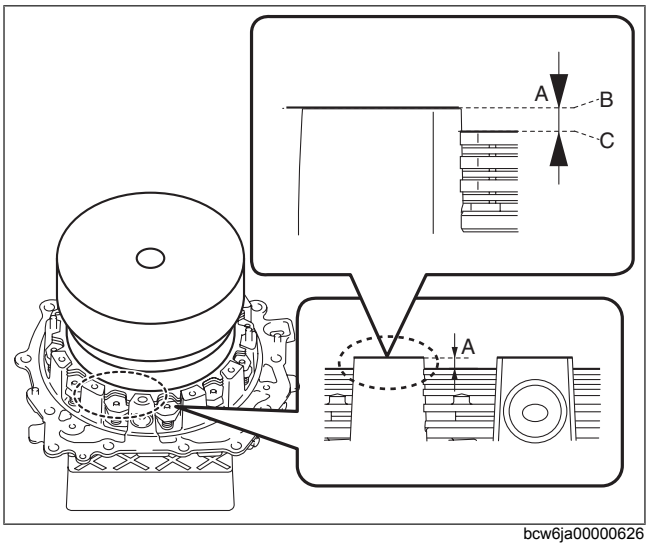
6. Measure distance B shown in the figure at the four locations (each separated by 120°) and calculate the average value of distance B.

Note

- Recommended measuring instrument: Depth micrometer
- Measure the splines of the retaining plate using a depth micrometer.

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A : Distance B
B : End cover end (alignment surface with brake housing)
C : Retaining plate end



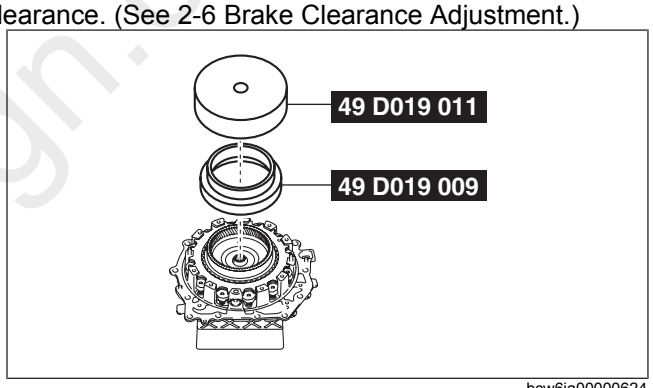
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7. Input the measured distance B and calculated distance B average value into the measurement/adjustment value input sheet.
8. Perform the following calculation to calculate the 2-6 brake clearance.
   2-6 Brake Clearance = E - C
   C: Average value of distance A
   E: Average value of distance B

   Note
   Example
   C: Average value of distance A is 3.150 mm {0.1240 in}
   E: Average value of distance B is 4.370 mm {0.1720 in}
   2-6 brake clearance = 4.370 mm {0.1720 in} - 3.150 mm {0.1240 in} = 1.220 mm {0.04803 in}
9. Input the calculated 2-6 brake clearance into the measurement/adjustment value input sheet.
10. Verify that the 2-6 brake clearance satisfies the specification.

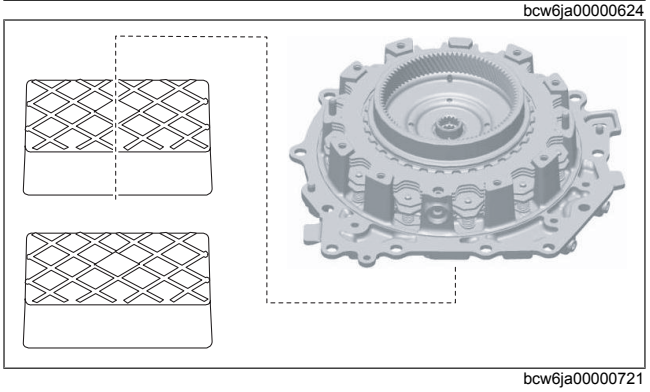
    Specification
    1.000—1.200 mm {0.03938—0.04724 in}

    • If not within the specification, adjust the 2-6 brake clearance. (See 2-6 Brake Clearance Adjustment.)
11. Remove the SSTs.



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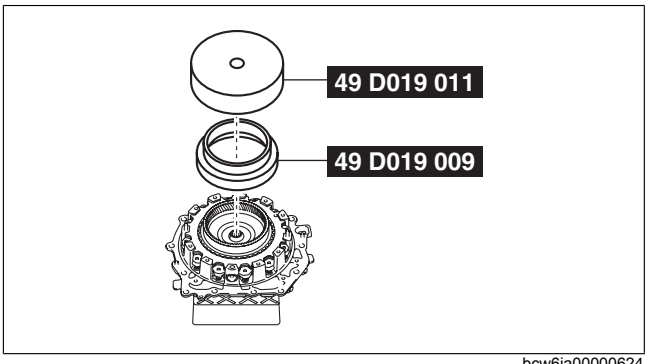
12. Take the end cover off the rubber plates.


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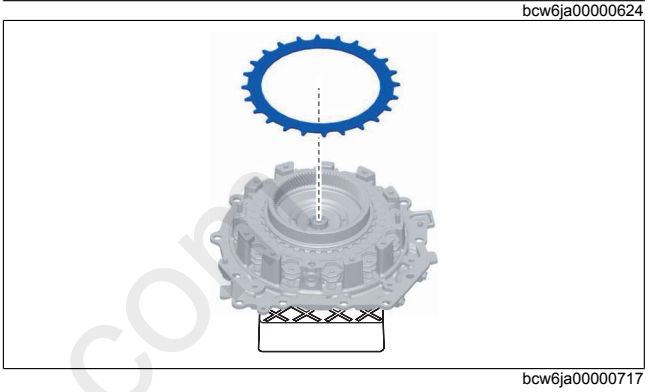
2-6 Brake Clearance Adjustment

1. Remove the SSTs.



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2. Remove the retaining plate.
3. Measure the thickness of the removed retaining plate.


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Note
- Recommended measuring instrument: Micrometer

4. Input the measured retaining plate thickness into the measurement/adjustment value input sheet.
5. Perform the following calculation to calculate the 2-6 brake clearance gap.

Note
- The 2-6 brake clearance gap is the difference between the 2-6 brake clearance and the median value of the 2-6 brake clearance specification.

2-6 brake clearance gap = F-J

F: 2-6 brake clearance

J: 2-6 brake clearance specification median value (1.100 mm {0.04331 in})

Note

Example

F: 2-6 brake clearance is 1.220 mm {0.04803 in}

2-6 brake clearance gap = 1.220 mm {0.04803 in} - 1.100 mm {0.04331 in} = 0.120 mm {0.00472 in}

6. Input the calculated 2-6 brake clearance gap into the measurement/adjustment value input sheet.
7. Perform the following calculation to calculate the optimum retaining plate thickness.

Thickness of optimum retaining plate = I + K

I: Thickness of removed retaining plate

K: 2-6 brake clearance gap

Note

Example

I: Thickness of removed retaining plate is 1.615 mm {0.06358 in}

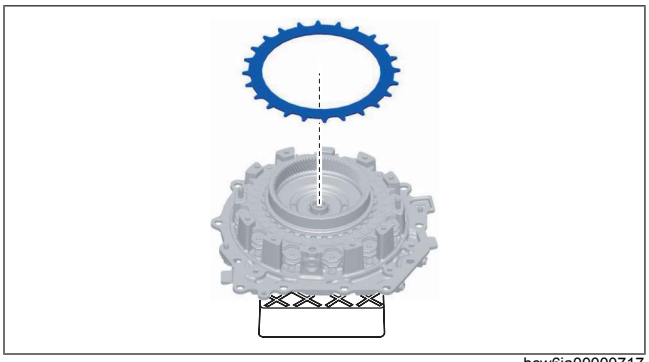
Thickness of optimum retaining plate = 1.615 mm {0.06358 in} + 0.120 mm {0.00472 in} = 1.735 mm {0.06831 in}

8. Input the calculated optimum retaining plate thickness into the measurement/adjustment value input sheet.
9. From the following table, select a retaining plate of a thickness closest to the calculated optimum thrust retaining plate.

Retaining plate thickness
2.2 mm {0.087 in}
2.1 mm {0.083 in}
2.0 mm {0.079 in}
1.9 mm {0.075 in}
1.8 mm {0.071 in}
1.7 mm {0.067 in}
1.6 mm {0.063 in}

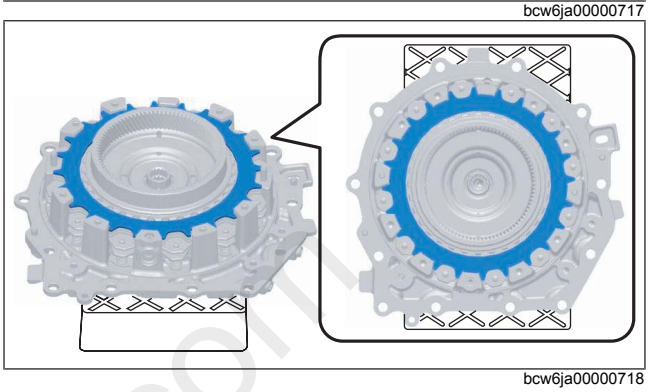
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10. Assemble the selected retaining plate.



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11. Perform the 2-6 brake clearance measurement from Step 5. (See 2-6 Brake Clearance Measurement.)



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